

RESTING MEMBRANE POTENTIAL & ACTION POTENTIAL

NotesGPT-Style High-Yield Summary

SOURCE SCOPE Built from the provided 39-slide membrane-potential/action-potential deck. No textbook file was attached in this request, so no textbook-only material has been added.

Organization: follows the slide sequence from ion distribution -> resting membrane potential -> local/action potentials -> ion-channel gating and propagation -> cardiac and smooth-muscle action potentials.

1. Ionic Distribution & Resting Membrane Potential

Slides 2-14: the electrochemical foundation of membrane voltage

Unequal ion distribution across extracellular fluid (ECF) and intracellular fluid (ICF)

- **Notation used throughout these notes:** ECF = fluid outside the cell; ICF = fluid inside the cell; **Na⁺** = sodium ion, **K⁺** = potassium ion, **Cl⁻** = chloride ion, **Ca⁺⁺** = calcium ion. The + or - sign shows the ion's electrical charge.

WHY: These abbreviations appear repeatedly in the slides, so defining them first makes every later equation and membrane-voltage diagram easier to read.

- The concentration table uses values such as **mEq/L**. For this deck, treat these as the slide's concentration values used to compare how much of an ion is present inside versus outside the cell.

WHY: The exam-relevant point here is the large inside-versus-outside difference, not a separate calculation with the concentration unit.

- **Extracellular fluid (ECF)** contains a large amount of **Na⁺** and only a small amount of **K⁺**; **intracellular fluid (ICF)** shows the opposite pattern.

WHY: A membrane voltage can only arise if ions are not distributed equally on the two sides of the membrane.

- **Cl⁻** is concentrated mainly in ECF, while **phosphates and proteins** are much more concentrated in ICF.

WHY: These unequal distributions create different chemical environments on the two sides of the membrane.

- The slide comparison gives representative concentrations such as **Na⁺ 142 mEq/L outside vs 10 mEq/L inside**, **K⁺ 4 vs 140 mEq/L**, and **Cl⁻ 103 vs 4 mEq/L**.

WHY: The large Na⁺ and K⁺ gradients make these ions especially important when the membrane becomes permeable to them.

- The figure also emphasizes very low intracellular **Ca⁺⁺** relative to extracellular Ca⁺⁺ and higher intracellular **phosphate/protein** content.

WHY: The figure is meant to show that each ion has its own concentration gradient rather than all solutes following one common pattern.

Species	ECF (slide figure)	ICF (slide figure)
Na ⁺	142 mEq/L	10 mEq/L
K ⁺	4 mEq/L	140 mEq/L
Ca ⁺⁺	2.4 mEq/L	0.0001 mEq/L
Mg ⁺⁺	1.2 mEq/L	58 mEq/L
Cl ⁻	103 mEq/L	4 mEq/L
HCO ₃ ⁻	28 mEq/L	10 mEq/L
Phosphates	4 mEq/L	75 mEq/L
Proteins	2 g/dL (5 mEq/L)	16 g/dL (40 mEq/L)

Definition and range of resting membrane potential (RMP)

- **Membrane potential** means the electrical voltage difference between the two sides of the cell membrane. **Resting** means the cell is not currently producing an action potential.

WHY: So an RMP of -70 mV means the inside is electrically negative compared with the outside at rest.

- **mV = millivolts**, the voltage unit used on the slide graphs.

WHY: The negative sign is part of the value and indicates which side is negative relative to the other.

- **Resting membrane potential** is the potential difference across the membrane of an **inactive cell**, with the **inside negative relative to the outside**.

WHY: RMP describes the resting electrical separation of charge across the membrane, not active signaling.

- RMP ranges roughly from **-10 to -100 mV** depending on the cell. The slides list a typical **neuron at -70 mV**, **skeletal muscle at -90 mV**, and a typical **epithelial cell at less than -50 mV**.

WHY: Different cell types have different membrane permeabilities and ion distributions, so they settle at different resting voltages.

Two requirements for a membrane potential

- A **concentration gradient** simply means an ion is present in different amounts on the two sides of the membrane. **Permeability** means the membrane provides a route through which that ion can cross.

WHY: The gradient creates a tendency to move; permeability determines whether that movement can actually occur.

- A potential difference requires **(1) an unequal distribution of one or more ions across the membrane** and **(2) membrane permeability to one or more of those ions**.

WHY: A concentration gradient provides the driving force, while permeability gives ions a path to move and separate charge.

Nernst potential and Nernst equation

- **Diffusion** here means ions moving because they are unequally distributed. A **diffusion potential** is the voltage produced as that ion movement separates electrical charge across the membrane.

WHY: The Nernst potential is the special voltage at which the electrical effect exactly counterbalances that ion's tendency to keep diffusing.

- In the equation, **[ion]_{in}** and **[ion]_{out}** mean the ion concentration inside and outside the cell. The slide also shows the more general symbols $E = RT/zF \log(...)$, but it does not define each constant; for this deck, the simplified 37 C form is the one the slides actually work through.

WHY: This keeps the equation tied to what the slide expects you to calculate and memorize rather than adding unsupported detail about constants that the deck never explains.

- The **Nernst potential (equilibrium potential)** for an ion is the diffusion potential that **exactly opposes net diffusion** of that ion through the membrane.

WHY: At this voltage, the electrical force opposing ion movement balances the concentration-driven tendency for that ion to diffuse.

- The general slide equation is $E = (RT / zF) \log([ion]_o / [ion]_i)$. For the simplified 37 C form shown, monovalent ions use **$\pm 61 \log([in]/[out])$ mV**; for **Ca⁺⁺**, the factor is **61/2**.

WHY: The charge on the ion changes how strongly voltage affects it, which is why divalent Ca⁺⁺ uses half the monovalent factor.

- The slide instructs: use **- for cations** and **+ for anions** in the simplified $[in]/[out]$ form.

WHY: The sign convention accounts for whether positive or negative charge is being moved across the membrane.

- Memorized equilibrium potentials given in the deck: **Na⁺ about +70 mV**, **K⁺ about -94 mV**, and **Ca⁺⁺ about +132 mV**.

WHY: These values show the membrane voltage toward which each ion would drive the cell if that ion dominated membrane permeability.

Ionic basis and origin of normal RMP

- **Na⁺ is actively transported out of cells and K⁺ into cells**, but at rest there are **more K⁺ channels**, so **K⁺ permeability is greater than Na⁺ permeability**.

WHY: Because K⁺ has the larger resting permeability, K⁺ movement dominates the resting voltage.

- The deck separates the normal RMP into contributions from **K⁺ diffusion**, **Na⁺ diffusion**, and the **Na⁺-K⁺ pump**.

WHY: RMP is not a single-process phenomenon; diffusion through channels and active transport contribute in different ways.

- K⁺ and Na⁺ diffusion alone produce about **-86 mV**, with almost all of that determined by K⁺ diffusion; the continuously acting **Na⁺-K⁺ pump adds about -4 mV**, giving about **-90 mV**.

WHY: This shows why K⁺ diffusion is the major immediate determinant, while the pump makes a smaller direct electrical contribution.

- The slide experiment with **ouabain**, an inhibitor of the Na⁺-K⁺ pump, showed that membrane potential can persist for **a few hours even when the pump stops**.

WHY: If the pump directly created the entire membrane voltage, blocking it would abolish the voltage immediately; persistence shows that stored ion gradients and membrane permeability are central.

- The **Na⁺-K⁺ pump is nevertheless essential for maintaining the K⁺ and Na⁺ concentration gradients** that support RMP over time.

WHY: The channels use existing gradients to generate voltage, while the pump preserves those gradients so they do not dissipate.

Goldman-Hodgkin-Katz (GHK) equation: combining multiple permeant ions

- The key new symbol is **P**, meaning the membrane's permeability to an ion. Thus **P_{Na}** is Na⁺ permeability and **P_K** is K⁺ permeability. **V_m** means membrane potential.

WHY: GHK asks not only "how much ion is on each side?" but also "how easily can each ion cross?"

- A **permeant ion** is simply an ion that can cross the membrane under the conditions being considered.

WHY: Only ions that can cross meaningfully influence the membrane voltage in this type of calculation.

- When both **Na⁺ and K⁺ are permeant**, membrane potential can be approximated by the **GHK equation**, which weights the concentrations of each ion by its **membrane permeability**.

WHY: Unlike the Nernst equation for one ion, GHK combines the influence of more than one permeant ion.

- The slide example states **P_K is 25 times P_{Na}**, so **P_{Na}/P_K = 0.04**; substituting the shown Na⁺ and K⁺ concentrations gives **V_m = -70.6 mV**.

WHY: Because K⁺ is much more permeable than Na⁺, the combined membrane voltage lies much closer to the K⁺ equilibrium tendency than to the Na⁺ tendency.

CORE RMP LOGIC High intracellular K⁺ + a membrane that is much more permeable to K⁺ at rest -> K⁺ leaves through K⁺ channels while intracellular negatively charged proteins cannot follow -> the inside becomes negative. The Na⁺-K⁺ pump maintains the gradients and contributes a smaller direct electrical effect.

Concept	High-yield point	Slide-supported number/example
RMP	Inside negative relative to outside	Neuron -70 mV; skeletal muscle -90 mV
Nernst potential	Voltage that exactly opposes net diffusion of one ion	Na ⁺ +70; K ⁺ -94; Ca ⁺⁺ +132 mV
Resting permeability	K ⁺ permeability > Na ⁺ permeability	P _K ≈ 25 x P _{Na} in GHK example
Diffusion contribution	Mostly K ⁺ -determined	K ⁺ + Na ⁺ diffusion ≈ -86 mV
Na ⁺ -K ⁺ pump	Maintains gradients; small direct voltage effect	≈ -4 mV direct contribution

2. Excitability, Local Potentials & Nerve Action Potentials

Slides 15-24: how excitable cells transition from resting voltage to an action potential

Excitable cells and types of excitation

- **Excitability** means the ability of a cell to change its membrane potential in response to a stimulus. A **stimulus** is the electrical, chemical, or mechanical input that starts that change.
WHY: The slides use excitability as the bridge between a stable resting voltage and a temporary signaling voltage change.
- **Excitable cells** listed in the slides include **neurons, muscle cells, some endocrine cells, and some other cells such as macrophages**. Electrical, chemical, or mechanical stimulation can change their membrane potential dramatically for short periods (milliseconds).
WHY: Excitability means the membrane voltage can respond rapidly to a stimulus instead of remaining fixed at its resting value.
- Two broad forms of excitation are shown: **local, non-propagated potentials** (synaptic, generator, or electrotonic potentials) and **propagated potentials (action potentials)**.
WHY: The key distinction is whether the voltage change stays local or regenerates along the membrane.

Local potentials

- A **local potential** is a membrane-voltage change that stays local rather than regenerating as a traveling action potential. The slide also shows that it can be **depolarizing** or **hyperpolarizing**.
WHY: "Local" tells you the voltage change is limited and can fade with distance.
- **Depolarization** means the membrane becomes less negative than its resting value; **hyperpolarization** means it becomes more negative.
WHY: These words describe the direction of the voltage change, not a separate structure or process.
- **Temporal summation** = local potentials add because they occur close together in time. **Spatial summation** = local potentials from different locations add together.
WHY: Both are ways for several smaller voltage changes to combine into a larger total change, as shown in the slide tracing.
- Local potentials can produce **depolarization or hyperpolarization**, and their size can vary with stimulus strength.
WHY: Local responses are graded rather than fixed in size.
- Local potentials can be **summed by temporal summation and spatial summation**.
WHY: Multiple local voltage changes can add together either across time or from different locations.
- The slide graph shows local potentials decreasing with distance from the stimulus site, in contrast with the non-decremental propagation later emphasized for action potentials.
WHY: A local potential is not regenerated at each segment of membrane, so its size can fade as it spreads.

Basic nerve action potential sequence

- An **action potential (AP)** is the propagated, rapid membrane-voltage event shown in the slide: resting negative voltage -> depolarization -> repolarization back toward the negative resting level.
WHY: Unlike a local potential, the AP is the full traveling electrical event.
- **Repolarization** means returning toward the resting negative membrane potential after depolarization. **Overshoot** means the tracing rises beyond 0 mV so the inside becomes positive relative to the outside.
WHY: These labels let you read the shape of the AP graph without treating each term as a separate unexplained concept.

- A nerve **action potential (AP)** begins with a sudden shift from the normal negative resting potential toward a positive potential (**depolarization**), followed by return toward the negative potential (**repolarization**).

WHY: The AP is a rapid, transient reversal and restoration of membrane voltage.

- During **depolarization**, the membrane suddenly becomes highly permeable to Na^+ , and Na^+ influx neutralizes the negative resting state.

WHY: Na^+ entering the cell carries positive charge inward, driving membrane voltage upward.

- In large nerve fibers, Na^+ influx can produce an **overshoot beyond 0 mV**; the slide notes that in small neurons and many CNS neurons the voltage may reach only around zero.

WHY: The extent of inward positive charge determines whether the membrane becomes truly positive or merely loses its negative voltage.

- During **repolarization**, Na^+ channels begin to close and K^+ channels open more than normal; rapid K^+ diffusion outward re-establishes the negative membrane potential.

WHY: Stopping Na^+ entry removes the depolarizing drive while K^+ leaving carries positive charge out, returning the inside toward negativity.

After-potentials / hyperpolarization

- The **spike potential** is the sharp, fast up-and-down portion of the AP tracing. An **after-potential** is the slower voltage change that follows the main spike.

WHY: The slide uses these names to divide the tracing into the main AP event and the smaller voltage changes that occur afterward.

- The sharp rise and rapid fall form the neuronal **spike potential**.

WHY: This is the fast central event of the nerve action potential tracing.

- The slower fall at the end is labeled **after-depolarization (negative after-potential)** in the slide.

WHY: The tracing does not always return instantly to the exact resting level after the main spike.

- A small overshoot in the hyperpolarizing direction with a prolonged tracing is labeled **after-hyperpolarization (positive after-potential)**.

WHY: The membrane can transiently become more negative than its resting level before fully settling back.

Initiation and threshold

- **Threshold** is the critical membrane-voltage level at which the Na^+ -driven positive-feedback process becomes self-amplifying and a full AP begins.

WHY: Below threshold the response can stay small; once threshold is reached, Na^+ entry promotes still more Na^+ entry.

- **Positive feedback** means the initial change reinforces itself: depolarization opens voltage-gated Na^+ channels $\rightarrow$ Na^+ enters $\rightarrow$ the membrane depolarizes more $\rightarrow$ more Na^+ channels open.

WHY: That loop explains the sudden steep rise of the action potential rather than a slow proportional increase.

- AP initiation depends on a **positive-feedback cycle involving Na^+ channels**.

WHY: Once Na^+ entry depolarizes the membrane enough, that depolarization promotes further Na^+ channel opening and even more Na^+ entry.

- The slides define **threshold** as the point at which Na^+ entering becomes greater than K^+ leaving. A sudden rise of about **15-30 mV** is usually required, with threshold values shown in the approximate **-50 to -70 mV** range.

WHY: Threshold marks the transition from a small local response to a self-amplifying depolarization.

- The threshold-for-excitation graph emphasizes both **stimulus intensity and duration**: subthreshold stimuli produce only local responses, whereas a stimulus that reaches threshold triggers a full AP.

WHY: A stimulus must alter membrane voltage enough, for long enough, to start the regenerative Na^+ feedback loop.

Feedback diagrams from the slides

- **Na⁺ positive feedback:** depolarizing stimulus -> opening of voltage-gated Na⁺ channels -> increased PNa -> increased Na⁺ influx -> further membrane depolarization.

WHY: Each step reinforces the next, so the depolarization accelerates once threshold is crossed.

- The Na⁺ positive-feedback loop is stopped by **inactivation of Na⁺ channels**.

WHY: Without termination, the regenerative cycle would not end and repolarization could not proceed.

- **K⁺ negative feedback:** Na⁺-driven depolarization -> opening of voltage-gated K⁺ channels -> increased PK -> increased K⁺ efflux -> repolarization.

WHY: K⁺ efflux reverses the voltage change that opened the K⁺ channels, so this loop counteracts depolarization rather than reinforcing it.

Feature	Local potential	Action potential
Propagation	Local / non-propagated	Propagated
Amplitude	Varies with stimulus	Triggered once threshold is exceeded
Summation	Temporal and spatial summation	Not described as summing in these slides
Distance	Can decline with distance	Propagates without decrement
Examples	Synaptic, generator, electrotonic	Nerve AP

3. Voltage-Gated Channels, Refractory Periods & Propagation

Slides 25-33: channel gating explains the AP shape, refractoriness, and conduction

Voltage-gated Na⁺ channel: activation and inactivation

- A **voltage-gated channel** is a membrane channel whose open/closed state changes when membrane voltage changes. A **gate** is the part of the channel shown opening or closing in the slide diagram.
WHY: This is why the membrane voltage itself can control ion permeability during an action potential.
- **Activation** = the Na⁺ channel opens rapidly and allows Na⁺ inward. **Inactivation** = a second gate closes more slowly, stopping Na⁺ flow even though the membrane has been depolarized.
WHY: The time delay between these two gates is what allows a brief Na⁺ influx rather than unlimited continued entry.
- When membrane potential reaches roughly **-70 to -50 mV**, the Na⁺ channel **activation gate opens rapidly**, allowing Na⁺ to pour inward.
WHY: The activation gate is the fast step that launches the rising phase once threshold is reached.
- The same voltage change also drives the **inactivation gate toward the closed state**, but this conformational change is **slower** than activation.
WHY: The delay allows Na⁺ influx to occur briefly before the channel becomes inactivated.
- The slide diagram shows Na⁺ channels cycling through **resting** → **activated** → **inactivated** states as membrane voltage rises and then returns.
WHY: The sequence explains why a voltage-gated Na⁺ channel can open quickly but then become temporarily unavailable even while the membrane is still depolarized.

Voltage-gated K⁺ channel

- For the K⁺ channel, **slow activation** means the channel opens later than the rapid Na⁺ activation described just before it.
WHY: The delayed timing helps explain why K⁺ outward movement becomes most important after the depolarizing upstroke has already started.
- As membrane potential rises from about **-90 mV toward zero**, voltage-gated K⁺ channels undergo a **slow conformational opening** that increases K⁺ diffusion outward.
WHY: The slower timing means K⁺ current becomes prominent after Na⁺-driven depolarization has already begun.
- K⁺ channel opening occurs at about the same time as **Na⁺ channel inactivation**.
WHY: The depolarizing Na⁺ current is therefore being shut down while the repolarizing K⁺ current is increasing.

Re-establishing Na⁺ and K⁺ gradients

- The **Na⁺-K⁺ pump** re-establishes the normal Na⁺ and K⁺ ionic gradients after activity; the slide states its activity may increase **up to 8-fold after an AP**.
WHY: The AP moves only a fraction of the available ions, but repeated activity still requires active transport to preserve the long-term gradients.
- The pump diagram shows **3 Na⁺ moved out and 2 K⁺ moved in** using ATP.
WHY: This is the transport pattern used by the membrane to restore and maintain the normal Na⁺/K⁺ distribution.

Absolute and relative refractory periods

- **Refractory period** means the interval after an AP when the membrane cannot fire normally again. **Absolute refractory period** = another AP cannot be triggered at all; **relative refractory period** = another AP is possible, but only with a stronger-than-normal stimulus.
WHY: Think of refractoriness as the membrane's temporary recovery period after firing.

- The later cardiac comparison abbreviates these as **ARP = absolute refractory period** and **RRP = relative refractory period**.

WHY: Defining the abbreviations here prevents them from appearing later as unexplained labels.

- During the **absolute refractory period**, **no stimulus, regardless of strength, can excite the nerve**.

WHY: The membrane is temporarily unable to generate another AP during this interval.

- During the **relative refractory period**, a **stronger-than-normal stimulus** can trigger excitation.

WHY: Excitability has begun to recover but has not yet returned to the normal resting level.

- The integrated slide graph links refractoriness to Na^+/K^+ channel states: excitability falls to **zero** during the absolute refractory period and then **gradually increases** during the relative refractory period.

WHY: Recovery of channel availability occurs progressively rather than instantaneously.

- In the same graph, Na^+ **permeability rises sharply and falls early**, whereas K^+ **permeability rises later and remains elevated longer**, matching depolarization followed by repolarization/hyperpolarization.

WHY: The timing of each ion permeability change mirrors the direction of the membrane-voltage change it produces.

Core properties of the action potential

- **All-or-none** means that once threshold is reached, the membrane produces the full AP event rather than a smaller “partial AP.” **Without decrement** means the AP does not progressively shrink as it travels.

WHY: These two properties distinguish a propagated AP from the graded, fading local potentials shown earlier.

- Depolarization must **exceed threshold** to trigger an AP.

WHY: Subthreshold voltage changes remain local; crossing threshold starts the regenerative event.

- The AP is **all-or-none**.

WHY: Once threshold is reached, the AP is generated as a full event rather than as a partially sized spike.

- The AP **propagates without decrement**.

WHY: Each new membrane segment regenerates the AP rather than merely receiving a fading passive voltage change.

Continuous and saltatory conduction

- **Propagation/conduction** means the action potential travels along the membrane. **Continuous conduction** means successive neighboring membrane regions depolarize along the fiber.

WHY: The electrical event is regenerated as it advances rather than simply drifting passively down the axon.

- In the myelinated-fiber diagram, **myelin** covers segments of the axon and the **nodes of Ranvier** are the exposed gaps between those segments. **Saltatory conduction** is the pattern in which only the nodes depolarize.

WHY: The slide’s central point is that concentrating depolarization at the nodes increases transmission speed and reduces the amount of membrane that must undergo the full ionic event.

- In **continuous conduction**, the AP travels in **all directions away from the stimulus**, including along branches of a nerve fiber, until the membrane has depolarized.

WHY: Adjacent membrane regions are successively brought to threshold so the excitation spreads along the fiber.

- In **saltatory conduction**, depolarization occurs at the **nodes of Ranvier** rather than along every part of the myelinated axon.

WHY: The slide emphasizes that restricting active depolarization to nodes changes how the impulse advances along a myelinated fiber.

- Saltatory conduction **increases nerve-transmission velocity** and **conserves axonal energy** because **only the nodes depolarize**.

WHY: Fewer membrane regions undergo the full ionic current sequence, so transmission is both faster and less energetically demanding according to the slide.

AP phase / interval	Dominant slide event	Functional consequence
Threshold / start	Rapid voltage-gated Na ⁺ activation	Regenerative depolarization begins
Depolarization	Na ⁺ influx; PNa rises sharply	Membrane potential moves positive
Early repolarization	Na ⁺ inactivation + delayed K ⁺ opening	Depolarizing current falls while outward K ⁺ current rises
Repolarization / after-potential	K ⁺ efflux remains elevated	Membrane returns negative; may transiently hyperpolarize
Absolute refractory	Excitability = zero in integrated graph	No stimulus can trigger another AP
Relative refractory	Excitability gradually recovers	Stronger-than-normal stimulus can trigger an AP

INTEGRATED CHANNEL LOGIC Fast Na⁺ activation gives the steep upstroke. Slower Na⁺ inactivation plus delayed K⁺ activation terminate the upstroke and produce repolarization. The timing of channel recovery creates the absolute and relative refractory periods.

4. Cardiac Action Potentials

Slides 34-38: ventricular/Purkinje, nodal, and atrial electrical patterns

Ventricular myocardium / Purkinje fibers

- This section's waveform applies to the slide category **ventricular myocardium / Purkinje fibers**. The deck does not further define Purkinje-fiber anatomy, so the notes keep the label without adding outside anatomy.

WHY: The goal here is to learn the ionic phases of the waveform exactly as the slide presents them.

- The ventricular/Purkinje action potential is divided into **phases 0-4**.

WHY: The phase system organizes the sequential ionic events across one cardiac action potential.

- **Phase 0: rapid Na⁺ influx.**

WHY: Rapid inward positive current produces the steep depolarizing upstroke shown in the tracing.

- **Phase 1: described as a small/insignificant phase with K⁺ out.**

WHY: The slide treats this as a brief transition before the plateau.

- **Phase 2 (plateau): Ca⁺⁺ in while K⁺ moves out.**

WHY: Opposing inward Ca⁺⁺ and outward K⁺ currents are shown during the prolonged plateau portion of the tracing.

- **Phase 3: rapid repolarization with K⁺ out.**

WHY: Outward K⁺ movement is the ionic event emphasized for returning the membrane toward its resting level.

- **Phase 4: ions are returned to the proper side between beats; the slide lists the ATPase pump, Na⁺-Ca⁺⁺ exchanger, and Ca⁺⁺ pump.**

WHY: The interval between action potentials is used to restore the ion distributions required for the next cycle.

Nodal tissue: the slow, self-activating action potential

- **Nodal tissue** is the slide's label for the cardiac tissue with the "slow" self-activating action potential. **Diastolic depolarization** means the membrane voltage gradually becomes less negative between action potentials until it approaches threshold.

WHY: That gradual rise explains the slide's statement that nodal tissue can activate itself repeatedly.

- **Autorhythmicity** means this self-activating, repeating electrical behavior. The slide attributes it to Na⁺ leak at rest and a membrane voltage that is already relatively close to threshold.

WHY: So "autorhythmic" is not a separate mysterious mechanism here; it is the result of the spontaneous drift toward threshold shown on the graph.

- Nodal tissue has a **"slow" action potential** and is **self-activating through diastolic depolarization**.

WHY: The membrane does not remain at one flat resting voltage; it drifts toward threshold on its own.

- The slides attribute diastolic depolarization to being **leaky to Na⁺ at rest**, producing **autorhythmicity**.

WHY: A gradual inward positive current moves the membrane toward the threshold for the next action potential.

- The slide shows nodal membrane potential already around **-70 mV** (with a note that Guyton gives **-55 mV**) and states gates open fully around **-40 mV**.

WHY: Because the starting voltage is closer to threshold than the -90 mV ventricular level, less depolarization is needed to reach the firing threshold shown.

- **Phase 0 is mostly Ca⁺⁺ influx**, so the upstroke is **slow**.

WHY: The slide directly links the dominant inward ion in phase 0 with the slower nodal depolarization.

- **Phase 3 is K⁺ efflux**, as in skeletal and nerve tissue.

WHY: Outward K⁺ movement again provides the repolarizing current.

Atrial action potential

- The **atrial action potential** is very similar to the **ventricular myocardial action potential**.

WHY: Atria and ventricles use a comparable phase pattern in the figure.

- The atrial AP is **smaller and shorter in duration**, with a **shorter refractory period**, especially a shorter **phase 2**.

WHY: The figure highlights a reduced plateau and earlier completion of the atrial waveform compared with the ventricular tracing.

Electrical vs mechanical timing: skeletal vs cardiac muscle

- **Electrical activity** refers to the action-potential tracing; **mechanical activity/response** refers to the muscle-tension tracing. The figure places them on the same time axis so their durations can be compared.

WHY: Reading both curves together is the point of the slide: skeletal and cardiac muscle differ in how much the electrical and mechanical events overlap.

- In the slide comparison, the **skeletal-muscle action potential is very brief** compared with the longer mechanical tension response.

WHY: The electrical event ends early while the plotted contraction continues for much longer.

- In cardiac muscle, the **action potential is prolonged**, and the **absolute refractory period (ARP)** occupies most of the time during which the mechanical response develops; a shorter **relative refractory period (RRP)** follows.

WHY: The cardiac electrical and mechanical events overlap much more extensively than the skeletal tracing in the provided figure.

Tissue / phase pattern	Key ionic or timing feature in slides	Distinctive point
Ventricle / Purkinje	0 Na ⁺ in; 1 K ⁺ out; 2 Ca ⁺⁺ in + K ⁺ out; 3 K ⁺ out; 4 ion restoration	Long plateau phase
Nodal tissue	Diastolic depolarization; phase 0 mostly Ca ⁺⁺ ; phase 3 K ⁺ out	Slow, self-activating / autorhythmic
Atrium	Similar to ventricle	Smaller, shorter AP; shorter phase 2 and refractory period
Skeletal muscle timing	Brief AP compared with contraction	Electrical spike ends well before mechanical response ends
Cardiac muscle timing	Prolonged AP + long ARP	Electrical/refractory period overlaps most mechanical response

5. Smooth-Muscle Action-Potential Patterns

- A **slow wave** in the final slide is the gradual, rhythmic change in membrane voltage on which repetitive spikes can appear. The slide specifically shows this in intestinal-wall smooth muscle.

WHY: The slow wave is the broad repeating baseline change; the narrow spikes are the actual action potentials drawn on top of it.

- A **plateau action potential** is an AP in which the membrane stays depolarized for a sustained interval before repolarizing, as shown for uterine smooth muscle.

WHY: "Plateau" refers directly to the long, relatively flat depolarized part of the tracing.

Slide 39: three electrical patterns shown in smooth muscle

- **A. Typical smooth-muscle spike potential:** a single action potential elicited by an **external stimulus**.

WHY: This tracing shows a discrete stimulus-triggered electrical event.

- **B. Repetitive spike potentials:** elicited by **slow rhythmical electrical waves** that occur spontaneously in smooth muscle of the **intestinal wall**.

WHY: The slow-wave baseline periodically reaches levels that trigger bursts of spikes in the figure.

- **C. Action potential with a plateau:** recorded from a smooth-muscle fiber of the **uterus**.

WHY: The tracing remains depolarized for a sustained interval rather than returning immediately after a narrow spike.

Pattern	Trigger / setting shown	Tracing appearance
Typical spike	External stimulus	Single brief spike
Repetitive spikes	Spontaneous slow rhythmical waves in intestinal wall	Bursts of spikes riding on slow waves
Plateau AP	Uterine smooth-muscle fiber	Prolonged depolarized plateau before repolarization

Rapid Final Review

WHY: The review below uses shorthand only for terms already explained in full above; use the main sections when a term is not yet automatic.

A slide-order consolidation for memorization

- **RMP needs two things:** an ion concentration gradient + permeability to that ion.

WHY: A gradient without a pathway cannot move charge; a pathway without a gradient has no net chemical driving force.

- At rest, **K⁺ permeability dominates Na⁺ permeability**, so K⁺ diffusion is the major determinant of RMP.

WHY: The membrane voltage is pulled most strongly toward the equilibrium tendency of the ion with the largest permeability.

- The **Na⁺-K⁺ pump maintains Na⁺/K⁺ gradients** and adds a smaller direct negative contribution to RMP.

WHY: The ouabain experiment shows the existing membrane potential persists temporarily even when the pump stops.

- **Na⁺ activation is fast, Na⁺ inactivation is delayed, and K⁺ activation is slow/delayed.**

WHY: That timing creates the upstroke, terminates Na⁺ entry, and drives repolarization.

- **Absolute refractory:** impossible to excite. **Relative refractory:** stronger-than-normal stimulus can excite.

WHY: Excitability first falls to zero and then progressively recovers.

- Action potentials are **threshold-dependent, all-or-none, and propagate without decrement.**

WHY: A regenerated AP behaves differently from a graded local potential.

- **Saltatory conduction** increases velocity and conserves energy because only **nodes** depolarize.

WHY: The myelinated axon does not need the full ionic depolarization sequence at every membrane segment.

- Ventricular/Purkinje cardiac AP: **0 Na⁺ in -> 1 K⁺ out -> 2 Ca⁺⁺ in + K⁺ out -> 3 K⁺ out -> 4 ion restoration.**

WHY: The phase sequence summarizes the ionic basis of the ventricular cardiac tracing.

- Nodal AP: **diastolic depolarization**, phase 0 mainly **Ca⁺⁺ in**, phase 3 **K⁺ out**.

WHY: This pattern explains the slow, self-activating nodal waveform shown in the slides.

- Smooth muscle may show a **single spike, repetitive spikes on slow waves, or a plateau AP.**

WHY: The final slide emphasizes that smooth-muscle electrical behavior is not represented by one universal waveform.

[End of Notes]